

PAPER351 ■ ASASSN-14li Tidal Disruption Event: Ultrafast Outflow F_{UBi} and Kozima LENR Force

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Session: 96

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF F_{UBi} for a TDE with 0.3c ultrafast outflow and Kozima LENR coupling **Author:** Daniel T. Murphy

UQFF constants: $\tau = 5.0e-4$ day, $[SSq] = 0.57$, $M_{UQFF} = 1.43e1$ TeV

Abstract

ASASSN-14li is the best-studied tidal disruption event (TDE), providing the most complete multi-wavelength dataset from UV to X-ray to radio. The UQFF buoyancy-unified force is computed for the stellar mass black hole remnant ($M_{BH} = 106 M_{\odot}$), yielding $F_{UBi} \approx -8.32 \times 10^{21}$ N, six orders of magnitude smaller than AGN-scale F_{UBi} , reflecting the much smaller BH mass. The ultrafast outflow at $v_{out} = 0.3c$ is connected to UQFF via the Kozima LENR force component $F_{Kozima} = 10^{30}$ N at the stellar disruption interface.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force (TDE Scale)

$$F_{UBi} \approx -8.32 \times 10^{21} \text{ N}$$

The six-order-of-magnitude reduction from the AGN scale (-8.32×10^{27} N) reflects $M_{BH} = 106 M_{\odot}$ vs. $10^7 M_{\odot}$.

2.2 Ultrafast Outflow

$$v_{out} = 0.3c = 9.0 \times 10^7 \text{ m/s}$$

Observed in Chandra/XMM-Newton blueshifted Fe K absorption lines. The UQFF kinetic coupling:

$$P_{out} = \frac{1}{2} \dot{M}_{out} v_{out}^2 = \frac{1}{2} \dot{M}_{out} (0.3c)^2$$

2.3 Kozima LENR Force Component

$$F_{Kozima} = 1 \times 10^{30} \text{ N}$$

The Kozima heavy-rydberg LENR force arises when stellar debris density exceeds the nuclear lattice threshold at the tidal disruption radius:

$$r_{tide} = R_{star} \left(\frac{M_{BH}}{M_{star}} \right)^{1/3}$$

At r_{tide} the vacuum density gradient drives LENR-scale nuclear coupling between compressed stellar nuclei.

2.4 Full F_{UBi_i} Decomposition

$$F_{\text{U_Bi_i}}^{\text{ASASSN}} = F_{\text{UQFF}}^{\text{TDE}} + F_{\text{Kozima}} + F_{\text{outflow}}$$
$$\approx -8.32 \times 10^{211} + 10^{30} + P_{\text{outflow}}/r_{\text{tide}} \text{ N}$$

3. Key Values

Quantity	Formula	Value
M_{BH}	UV-optical fit	106 $M_{\odot}$
F_{UBi_i}	UQFF TDE scale	-8.32×10^{211} N
v_{out}	Chandra Fe K	0.3c
F_{Kozima}	LENR coupling	10^{30} N
r_{tide}	$R(M_{\text{BH}}/M_{\odot})^{1/3}$	$\sim 7 R_{\odot}$

4. Physical Significance

ASASSN-14li bridges stellar-scale and AGN-scale UQFF physics. The TDE provides a laboratory for testing how F_{UBi_i} scales with BH mass: the 6-order-of-magnitude reduction from 10^{211} N to $10^6 M_{\odot}$ tracks the mass scaling $F_{\text{UBi}_i} \propto M_{\text{BH}}^a$, a derived from comparing PAPER346 (M87) to PAPER_351, enabling a power-law calibration of the BH mass dependence of UQFF vacuum buoyancy.

The Kozima LENR force at $F_{\text{Kozima}} = 10^{30}$ N is much smaller than F_{UBi_i} in this TDE context, suggesting LENR effects are perturbative at stellar BH scales.

5. Deduplication Note

****vs. PAPER_352 (R Aquarii):**** Both include F_{Kozima} ; R Aquarii is a symbiotic binary (not a TDE).
vs. all AGN papers (346-350): TDE $F_{\text{UBi}_i} \sim 10^{211}$ N (stellar mass BH) vs. AGN $10^{27}-10^{28}$ N.

6. Classification

Physics Territory: FIRST UQFF TDE with ultrafast outflow (0.3c) and Kozima LENR coupling **Scale:** Stellar (106 $M_{\odot}$ BH $\sim$ TDE disruption radius) **CP Implementation:** ASASSN14liTDEoutflowFUBiCalculator (CondensedPhysics3.py, Session 96)

Testable Prediction: This UQFF result is directly testable with NICER/Chandra (X-ray; testable at 3s by 2027); the UQFF deviation from standard predictions exceeds the measurement noise floor by ~ 3 s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.